

LR-CSW-51T2 + LR-NCE-A0 — ASIC Tape-Out Checklist

Issued: 2026-04-19 • **Process:** TSMC N3 (CoWoS-L for LR-CSW; FCBGA for LR-NCE) **Owner:** LightRail AI — Silicon Engineering

This checklist is the master release gate for foundry hand-off (GDSII / OASIS frame). Every row is signed below by the LightRail AI engineering function that owns the verification artefact.

1. RTL freeze

#	Item	Owner	Sign-off artefact	Status
1.1	RTL frozen at LR-DV/rtl/{common,lr_csw,lr_nce}/	Digital Design	git tag rtl-tapeout-v1	Approved
1.2	Coding-standard compliance	Digital Design	lint/questa_lint/lint.errors.rpt — 0 errors	Approved
1.3	Synthesisable-subset audit	Digital Design	lint.warnings.rpt — 0 fatal	Approved

2. Functional verification

#	Item	Owner	Sign-off artefact	Status
2.1	UVM regression — 4 000 seeds clean	Verification	tb/cov/coverage_report.txt 100 % L/B/F/T/ Funcl	Approved
2.2	SVA assertions all proven	Verification	Questa formal lint/sva.rpt — 0 fail	Approved
2.3	Performance — 51.2 Tbps line rate	Verification	tb/perf/perf_51T.csv — 100 % LR @ 9 KB MTU	Approved

3. Static / formal sign-off

#	Item	Owner	Sign-off artefact	Status
3.1	Questa CDC — 0 violations	Verification	lint/questa_cdc/cdc.violations.rpt	Approved
3.2	Questa RDC — 0 violations	Verification	lint/questa_rdc/rdc.violations.rpt	Approved
3.3	PrimeTime setup/hold sign-off	Static Timing	signoff/sta/reports/setup.rpt WNS = 0 ns	Approved
3.4	Conformal LEC RTL↔netlist	Verification	signoff/formal/reports/compared.rpt clean	Approved

4. Physical sign-off

#	Item	Owner	Sign-off artefact	Status
4.1	Calibre DRC	Physical Design	signoff/drc_lvs/reports/drc.summary — 0 violations	Approved
4.2	Calibre LVS	Physical Design	signoff/drc_lvs/reports/lvs.report — clean	Approved

#	Item	Owner	Sign-off artefact	Status
4.3	Calibre antenna	Physical Design	signoff/drc_lvs/reports/antenna.rpt — clean	Approved
4.4	Voltus EM/IR (static + dynamic)	Power Integrity	signoff/power/reports/ir.*.rpt < 75 mV	Approved
4.5	PrimePower / Voltus average power	Power Integrity	signoff/power/reports/power.hier.rpt ≤ 350 W	Approved

5. DFT / production-test

#	Item	Owner	Sign-off artefact	Status
5.1	Scan coverage ≥ 99 %	DFT	dft/scan_coverage.rpt	Approved
5.2	MBIST + LBIST — full closure	DFT	dft/bist.rpt	Approved
5.3	JTAG 1149.1 + 1149.6 BSDL	DFT	dft/bsdl/lr_csw.bsdl	Approved

6. Low-power closure

#	Item	Owner	Sign-off artefact	Status
6.1	UPF 3.1 consistency	Low-Power	signoff/lp/upf_check.rpt	Approved
6.2	Isolation / retention strategy	Low-Power	signoff/lp/iso_retain.rpt	Approved
6.3	Power-state-transition coverage	Verification	tb/cov/pst_cov.rpt 100 %	Approved

7. Tape-out package

#	Item	Owner	Sign-off artefact	Status
7.1	GDSII frame + bias check	Physical Design	signoff/drc_lvs/reports/gds_frame.rpt	Approved
7.2	OASIS export (foundry preferred)	Physical Design	pnr/innovus/out/lr_csw_top.oas.gz	Approved
7.3	Layer map + density fill	Physical Design	signoff/drc_lvs/reports/density.rpt	Approved
7.4	Foundry data-room manifest	Programme Office	docs/Foundry_Manifest.csv	Approved

8. FPGA prototype gate (parallel to ASIC)

#	Item	Owner	Sign-off artefact	Status
8.1	Vivado build clean, WNS ≥ 0 ns	FPGA Engineering	fpga/vivado/out/timing_post_route.rpt	Approved
8.2	Bitstream + ILA probes	FPGA Engineering	fpga/vivado/out/lr_csw_fpga.bit / .ltx	Approved
8.3	4-FPGA partitioning closure	FPGA Engineering	docs/LR-DV_Digital_Design_Verification_FPGA_Plan.md §9	Approved

9. Approvals — release to foundry

Role	Approver	Date	Status
Digital design lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
Verification lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
Physical design lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
Static-timing lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
Power integrity lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
Low-power lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
DFT lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
FPGA prototype lead	LightRail AI — Silicon Eng.	2026-04-19	Approved
Hardware VP	LightRail AI — Hardware Eng.	2026-04-19	Approved
Programme office	LightRail AI — Programme	2026-04-19	Approved for data-room